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Digital Twin: New Back Allocation Algorithm Based on the Maximum Likelihood Principle

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Abstract

The rapid advancement of digital technologies in the oil and gas sector has led to the development of innovative solutions for optimizing production and enhancing asset management. One such solution is the Digital Twin, a dynamic, real-time digital representation of physical assets. This paper introduces a novel back allocation algorithm for gas and gas condensate fields based on the Maximum Likelihood Estimation (MLE) principle, integrated within the Digital Twin framework.

The core thesis of this paper is the application of the MLE principle to the back allocation of fluids to wells and reservoirs, solving this task with the highest possible likelihood. The MLE-based approach allows for precise determination of fluid origins, pinpointing the specific wells and even the intervals within those wells, including the detailed component content.

For the MLE algorithm to function correctly, robust fluid characterization is essential. This involves comprehensive compositional grading and the use of Equations of State (EOS) to model fluid properties accurately. Understanding the compositional distribution from the output to the input, along with knowledge of how composition changes with depth and phase behavior at every node in the system, is crucial for achieving accurate back allocation using MLE.

Key aspects of the Digital Twin and MLE-based back allocation algorithm include:

1. **Accurate Fluid Allocation:** Determining the most probable production rates and fluid origins by adjusting parameters such as flow rates, fluid compositions, and equipment dimensions to minimize deviations.
2. **Compositional Grading:** Utilizing detailed compositional data to understand how fluid properties change with depth, which is vital for accurate modeling and allocation.
3. **Equation of State (EOS) Models:** Employing EOS models to accurately describe fluid phase behavior under various conditions, providing the necessary data for the MLE algorithm to function effectively.

This integration of MLE within the Digital Twin environment enhances the accuracy of production allocation, leading to significant improvements in operational efficiency and economic performance. Case

studies from gas and gas condensate fields demonstrate the algorithm's capability to optimize production allocation, reduce operational risks, and enhance overall asset management.

This paper will detail the technical aspects of the MLE-based algorithm, the importance of fluid characterization and EOS models, and the resulting benefits in terms of production optimization and economic gains. The findings suggest that adopting this advanced back allocation approach can lead to substantial improvements in field development strategies and operational efficiency.

Keywords: Digital Twin, Maximum Likelihood Estimation, Back Allocation, Fluid Characterization, Equations of State, Compositional Grading, Oil and Gas Production, Operational Efficiency, Asset Management

Background/Introduction

The challenge of back allocation of gas and condensate production has been a longstanding issue, with no optimal solution currently available. The main difficulties arise from the fact that the quantitative accounting of commercial fluids is carried out at the field's export point, i.e., after the stage of gathering and processing of the fluids. Typically, fluids from dozens, hundreds, or even thousands of wells are collected at the gathering point (central processing facilities). Each well's fluid, collected from different reservoir depths, is a weighted average mixture of fluids from one or more productive horizons. Complicating matters further is the impracticality of using full-component reservoir fluid models for both reservoir modeling and surface processing due to the excessive computational time required for multi-scenario forecasting.

The lack of an acceptable solution to the inverse problem leads to incorrect allocation of initial hydrocarbon reserves, as the produced commercial fluid is incorrectly attributed across different parts of the reservoir laterally and vertically, whether from one or multiple productive horizons during joint reservoir exploitation. This disrupts the field's material balance and hampers reservoir development management.

Current technologies do not allow for the production accounting of each well individually. The use of multiphase metering in fields with hundreds or thousands of wells is economically unfeasible. Additionally, multiphase metering equipment requires permanent calibration based on laboratory analysis of sampled fluids. In gas-condensate fields with high C5+ content, the results of multiphase metering can significantly deviate from actual production rates due to the imperfect separation of the multiphase flow into different phases. This problem remains unresolved in cases where the hydrocarbon gas and liquid phases have relatively similar properties.

Gas-condensate well tests play a crucial role in solving the inverse problem. The maximum possible number of such tests, depending on the size and maturity of the field, can significantly reduce uncertainties and more accurately allocate commercial production. The main measured parameters here are fluid flow rates through a three-phase full-stream separator, bottomhole and wellhead pressures and temperatures, and the composition and properties of different phases of the produced mixture. While gas-condensate tests cannot be performed for all wells in the field, they allow for characterizing the reservoir fluid in the tested wells, reducing the number of possible solutions to the inverse problem without requiring labor-intensive PVT analyses of sampled fluids.

The results of flow rate measurements during field geophysical surveys of wells are crucial for solving the inverse problem, as they provide information on inflow depths and the amount of fluid entering the well. This data is vital for interpreting the composition of the produced fluid mixture from the well. In describing the complex fluid of large fields, such information is invaluable.

Reservoir fluid in the reservoir is a complex multicomponent system. The properties of the reservoir fluid are interconnected with its composition and follow physical laws. Studying the composition and properties of the reservoir fluid allows for building a high-quality model that can be used to solve not only traditional tasks of predicting fluid properties in the reservoir, but also modern tasks related to the creation and operation of a Digital Twin of the field.

Proposed Methodology

Changes in the composition and properties of the produced fluid, recorded in reservoir fluid samples, are related to several factors:

- Changes in thermobaric conditions at different depths within the productive formation.
- Changes in the composition of the reservoir fluid with depth, caused by different influences of gravitational forces, the gradient of reservoir temperature, and thermodiffusion on components of different masses within the fluid.
- The lack of hydrodynamic connectivity in one or more reservoirs or productive horizons (faults, reservoir pinch-outs, impermeable layers between formations, etc.), different sources of hydrocarbons formed from organic matter of different types and at different stages of catagenesis.
- Fluid depletion as reservoir pressure falls below the condensation onset pressure. Reservoir condensate losses and varying flow rates of gas and liquid phases directly affect the final composition of the produced fluids.

To describe the variations in the composition and properties of the reservoir fluid across productive horizons and formations within the studied field, reservoir fluid samples are collected and studied in the laboratory. During gas-condensate well tests, samples are taken for physical recombination of the reservoir fluid in the laboratory, and PVT experiments are conducted. The resulting information allows describing the properties and composition of the fluid under initial conditions and changes in thermobaric conditions during field development, both in the reservoir and in surface infrastructure.

Deep samples taken with a formation tester from specified reservoir depths, along with laboratory PVT studies of these samples, provide experimental data on changes in fluid composition and properties with depth. This data, in addition to similar tasks for recombined samples, helps to construct a mathematical model of the fluid and predict changes in fluid composition and properties under reservoir and surface conditions.

A mathematical model (EOS) of the reservoir fluid is typically built using a cubic three-parameter equation of state, modified by Soave-Redlich-Kwong or Peng-Robinson. The main components of the model are the composition and results of laboratory PVT experiments for each reservoir fluid sample. Since EOS-based property simulations cannot reproduce experimental PVT measurements with sufficient accuracy due to uncertainties in estimating fraction parameters (critical pressure and temperature, acentric factor, etc.), the EOS parameters require tuning. This tuning is performed using regression analysis, which ensures convergence of the simulated and experimental properties of the reservoir fluid across all quality samples within acceptable error margins.

Key Aspects of the Back Allocation Algorithm Based on the Digital Twin and MLE:

1. **Fluid Allocation:** Determining the most probable production rates and fluid origins by adjusting parameters such as flow rate, fluid composition, and equipment size to minimize deviations.
2. **Compositional Grading:** Using detailed composition data to understand how fluid properties change with depth, which is crucial for accurate modeling and allocation.
3. **Equations of State (EOS) Models:** Utilizing EOS models to accurately describe the phase behavior of fluids under various conditions, providing essential data for the effective operation of the MLE algorithm.

Integrating MLE into the Digital Twin environment enhances the accuracy of production allocation, leading to improved operational efficiency and economic performance.

Results

For reservoir gas samples, one of the key results of laboratory PVT experiments is the liquid phase volume isotherms during pressure reduction in CCE (Constant Composition Expansion) and CVD (Constant Volume Depletion) experiments (Figure 1). Achieving convergence between the simulation results and experimental data for this parameter across all samples allows for predicting reservoir fluid properties during reservoir depletion modeling.

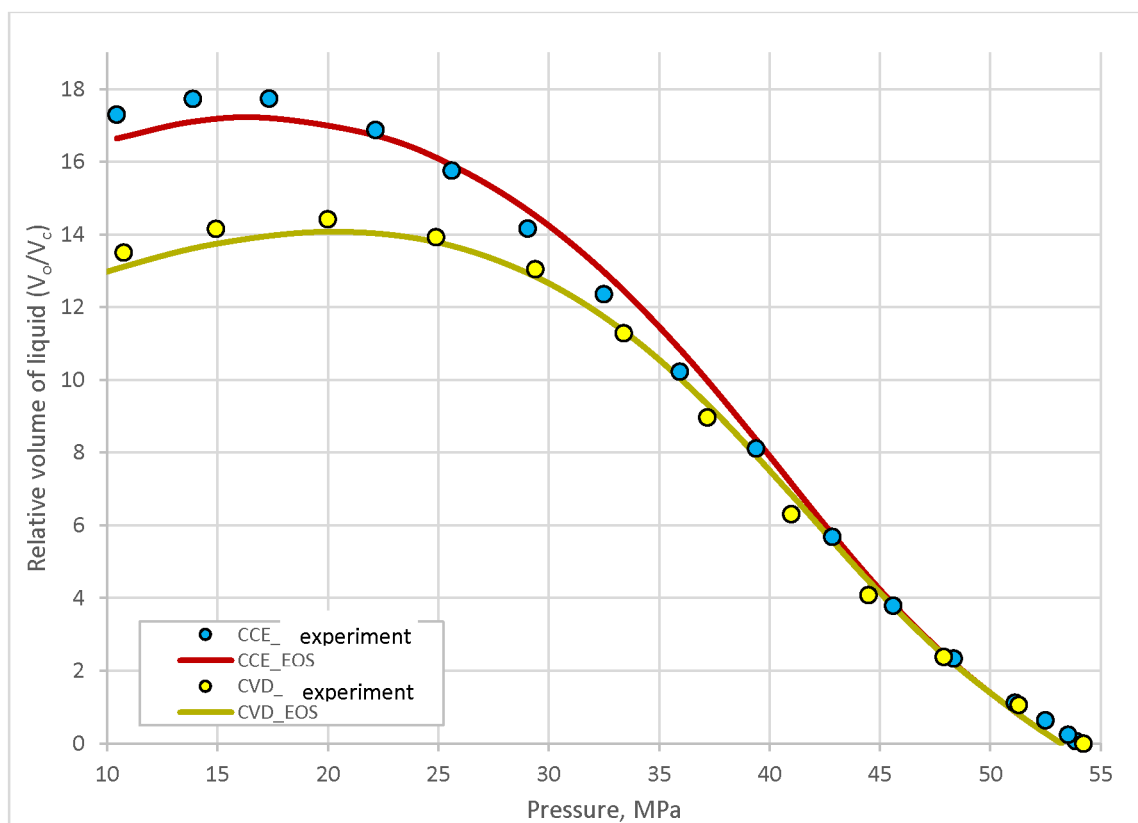


Figure 1—Comparison of calculated and experimental (CCE, CVD) properties of reservoir gas

Another experiment whose results must be compared with model calculations is stepwise separation conducted during field gas-condensate well tests. Here, the model must reproduce the phase volume ratios at the separator under specified conditions, as well as the properties and composition of the produced fluids. Stepwise separation allows tuning the equation to reproduce the composition, properties, and phase ratios of the fluids under surface conditions, i.e., at pressures and temperatures different from reservoir conditions.

To account for changes in reservoir fluid composition and properties occurring during secondary gas injection methods for enhanced recovery of the liquid phase from the reservoir, the initial fluid model needs adjustment based on the results of special PVT studies of the reservoir fluid.

The properties of the initial reservoir fluid change not only due to the direct influence of thermobaric conditions associated with different depths of fluid within the reservoir but also due to the differentiation of fluid components under the influence of gravitational forces, high reservoir temperature (thermodiffusion), and low permeability of the reservoir rock. The composition of reservoir fluid in a formation (productive formation) can vary slightly with depth, or it can differ significantly between the upper and lower parts of the same formation, directly affecting the composition of the produced fluids.

Once the EOS model is created, a fluid model describing the differentiation of composition with depth, known as Compositional Grading (CG), is required. This is important when there is a significant difference

in fluid composition and properties with depth, as observed in field (gas-condensate well tests) and laboratory (PVT) studies. As the accuracy requirements for reproducing the composition and properties of the produced fluid on the surface increase, the CG model becomes essential for solving the inverse problem in the Digital Twin of the field. Therefore, obtaining experimental data that allows assessing the composition gradient with depth is a priority task when studying the field.

The CG model is built based on the results of laboratory PVT analyses of deep samples taken from different depths of the formation using a formation tester, as well as a large number of composition measurements of the produced fluid from individual wells across the field.

The molar enthalpy plays a crucial role in creating the molar enthalpy change, which allows for accounting for the high compositional differentiation with depth present in some fluids, including reservoir gas, on which this study is based. Figure 2 illustrates the relationship between the C5+ component concentration in the reservoir gas and depth in a major gas-condensate field, as well as the CG model variations constructed to describe the compositional differentiation with depth.

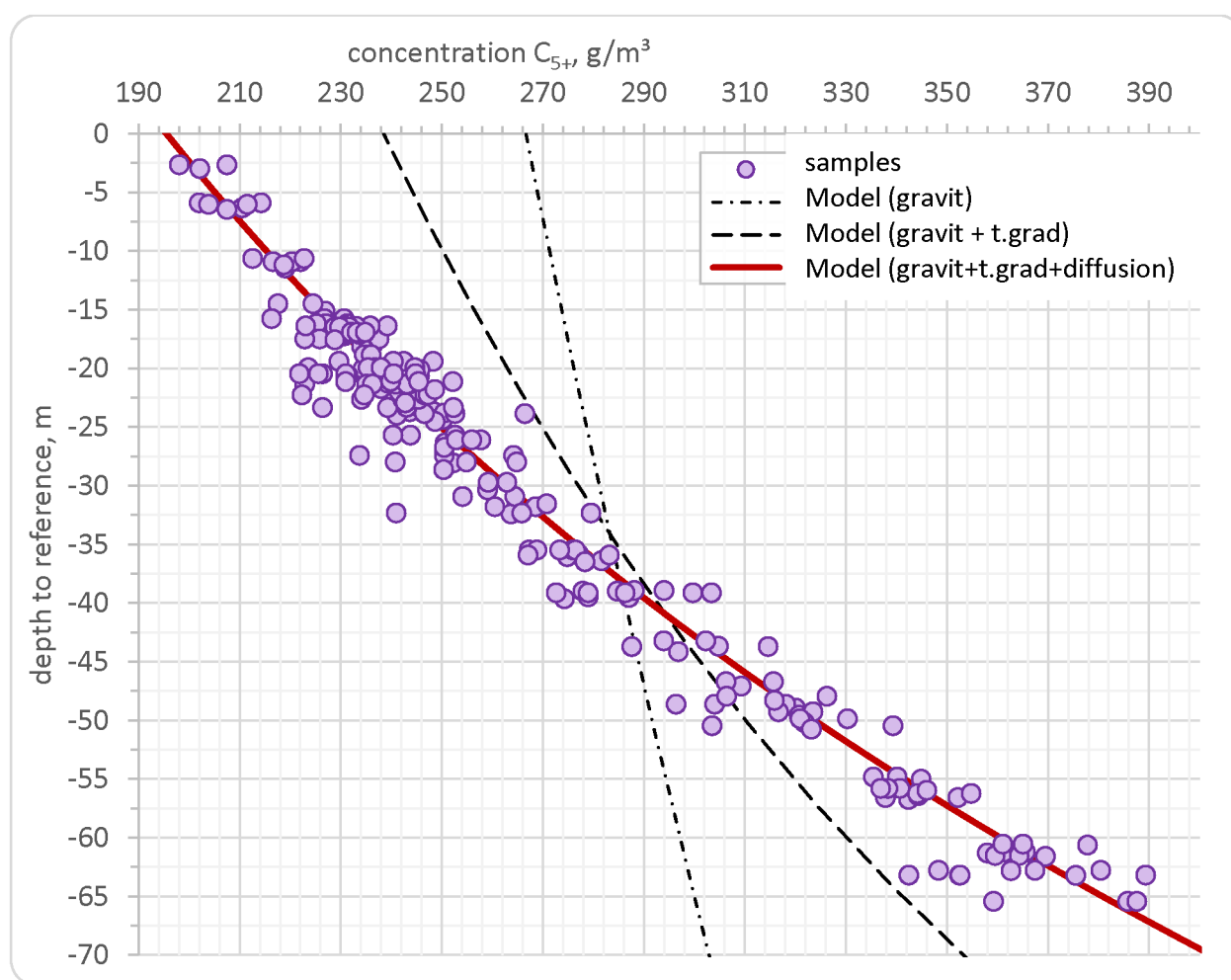


Figure 2—Concentration of C5+ Components vs. Depth

After creating the reservoir fluid model in the Digital Twin, data collection from pressure and temperature measurement sensors begins. These sensors provide continuous recording of the current thermobaric conditions in the field, located at the bottomhole and wellhead, and at various points in the field infrastructure (Figure 3)—pipelines, gathering points, gas processing facilities (GPP), and compressor

stations. The mass flow rates of the liquid and gas phases of commercial products, as well as their composition, are also continuously recorded using inline gas chromatographs.

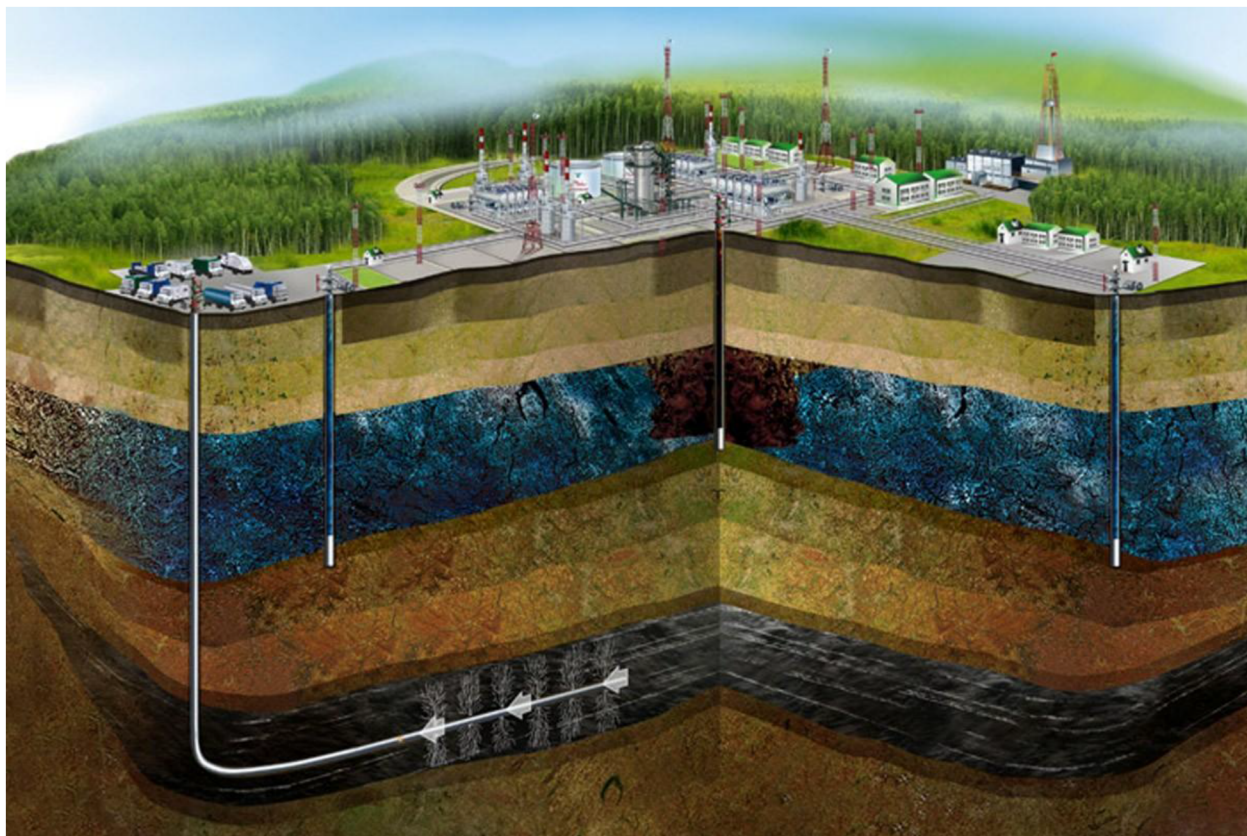


Figure 3—Schematic Representation of Field Gathering and Preparation of Commercial Products for Back Allocation of Hydrocarbon Fluids

In the presented case, the intervals of penetration of the productive formation reach hundreds of meters. Therefore, flow rate data from PLT surveys, with an assessment of fluid inflow depths and quantities, is of great importance. This allows for estimating the average depth of fluid inflow from the productive formation and more accurately distributing the well's fluid production.

A closed system that defines "composition-properties" at every point from the reservoir to the commercial product output adheres to material balance and can be solved using a system of phase equilibrium equations. The back allocation of commercial production, both by wells and by fluid inflow depths within the well, is then calculated.

For calculating the composition of the incoming mixture under conditions existing at measurement points, inverse two-phase or three-phase flash calculations are used, allowing the assessment of phase equilibrium compositions.

In a simplified scenario, consider a closed system example at the first-stage separator of reservoir gas (Figure 4).

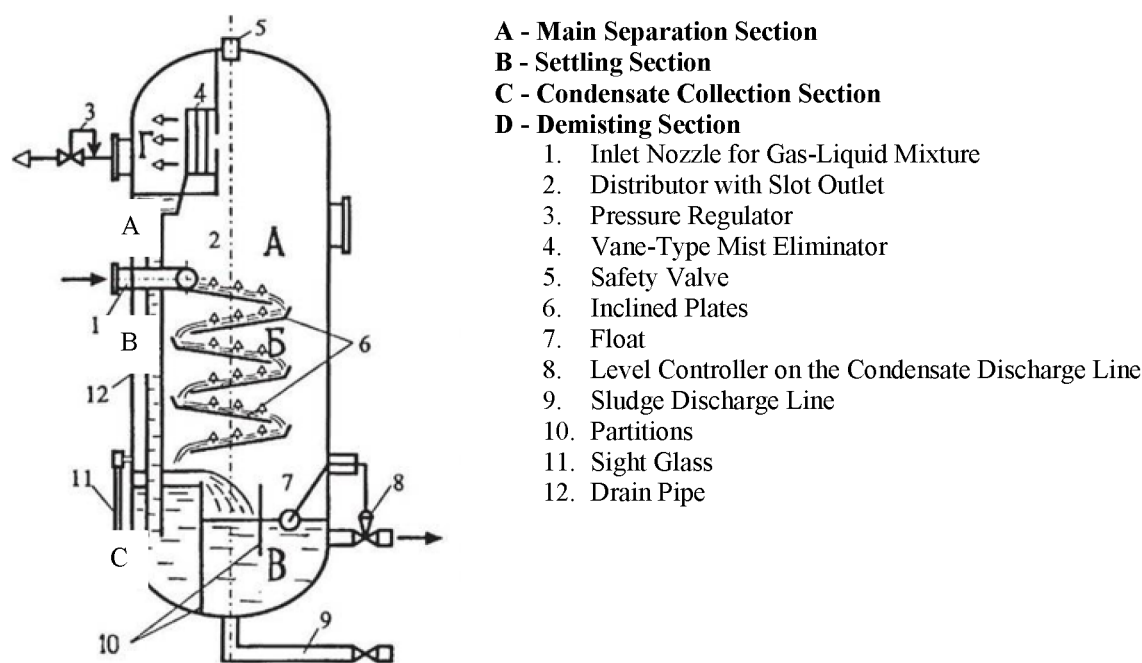


Figure 4—Diagram of a Vertical Separator and Controlled Nodes

The separator receives a mixture of produced fluid at known pressure and temperature. The thermobaric conditions, mass flow rates of the gas and liquid phases per unit time, and the amount of fluid entering the separator are known from material balance. The composition of the gas phase exiting the separator is determined almost continuously by an inline gas chromatograph, and the density of the condensate is known. The inverse problem of determining the composition of the mixture entering the separator is solved using the flash calculation algorithm proposed by Michelsen. Similar tasks are solved in other CPF elements. Where processes such as fluid flow through a valve, approximated as constant enthalpy, or through a compressor, approximated as constant entropy, are modeled, flash separation with appropriate variables allows solving the inverse problem of determining the fluid composition at the inlet.

The advantage of solving the back allocation problem within the Digital Twin, unlike hydrodynamic and integrated field models, is the ability to use the full-component reservoir fluid model for compositional calculations via the equation of state. This is possible due to the strategic horizon of predictive information used in the Digital Twin, typically limited to operational (hours, days) or monthly analysis and forecast periods. These features allow for the use of a full-component fluid model in the Digital Twin to solve business tasks of optimizing well and field equipment operation and ensuring uninterrupted production of liquid and gas hydrocarbons in real-time.

The best results in solving the back allocation problem by wells and fluid inflow depths were achieved using the statistical maximum likelihood estimation (MLE) method. The input experimental data for solving the inverse problem by MLE in the Digital Twin include:

- Compositional model (EOS) of reservoir fluid (initial information and adjustment with each model change);
- Depth compositional differentiation model (CG) of reservoir fluid (initial information and continuous updating in real-time);
- Detailed CPF separation scheme and elements with thermobaric conditions of produced fluid separation (initial information and adjustment with each scheme change);
- Field measurements of wellhead pressures and temperatures at various points in the field infrastructure (CPF) (continuous data recording);

- Measurements of production rates per unit time for each phase—gas, liquid hydrocarbons, and water (continuous data recording);
- Measurements of the produced fluid composition by inline chromatograph (continuous data recording);
- Recording information during gas-condensate well tests—wellhead and bottomhole pressures and temperatures, production rates, composition, and properties of the products (recorded as new data becomes available);
- Recording information on PVT test results of recombined or deep reservoir fluid samples (recorded as new data becomes available);
- Recording PLT information in wells, allowing for the assessment of inflow depths and the contribution of each fluid portion entering the well from the productive interval (recorded as new data becomes available).

Using the MLE method, the produced fluid is allocated by wells, and then, using the distribution of fluid composition by depth and flow rate data, by depth in reservoir conditions. The MLE method is described below.

The probability density that the value (a) will take on the values $(x_1 \pm \sigma_1)$, $(x_2 \pm \sigma_2)$, $(x_3 \pm \sigma_3)$, ..., where x and σ are the value and measurement error, respectively, is expressed by the likelihood function:

$$W(a) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left[-\frac{(x_i - a)^2}{2\sigma_i^2}\right]$$

The logarithmic form of the likelihood function is:

$$\ln W(a) = -\sum_{i=1}^n \frac{(x_i - a)^2}{2\sigma_i^2} + \sum_{i=1}^n \ln \frac{1}{\sqrt{2\pi\sigma_i^2}}$$

By determining the maximum of the likelihood function, the most probable value of the measured quantity a^* can be found. Differentiating the logarithmic function, we get:

$$\frac{\partial L}{\partial a} = \sum_{i=1}^n \frac{x_i - a}{\sigma_i^2}$$

Setting $\frac{\partial L}{\partial a}$ equal to zero, we get the following expression:

$$a^* = \frac{\sum_{i=1}^n \frac{x_i}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}}$$

This paper describes the technical aspects of the MLE-based algorithm, the importance of fluid characterization and EOS models, and the resulting benefits in terms of production optimization and economic gains. The results demonstrate that adopting an advanced back allocation approach can lead to significant improvements in field development strategies and operational efficiency.

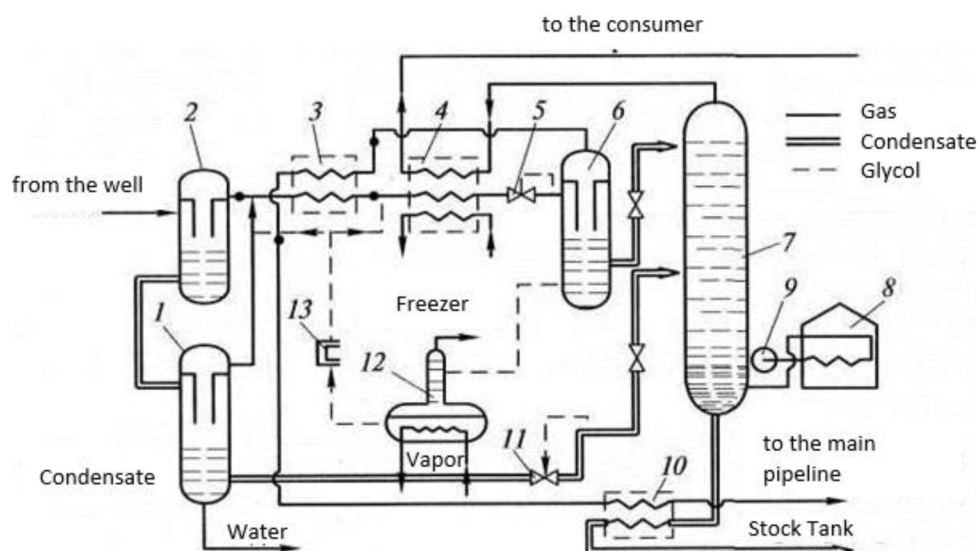


Figure 5—Scheme of the Low-Temperature Separation of Reservoir Gas (1, 2, 6 — separators, 3, 4, 10 — heat exchangers, 5 — regulator, 7 — stabilization column, 8 — reboiler, 9, 13 — pumps, 11 — throttle, 12 — regeneration plant)

Conclusions

Solving the inverse problem of allocating produced fluids by wells and depth allows solving the problem of production allocation when writing off initial hydrocarbon reserves and accounting for production according to mass material balance. Dynamic model correction and updates to geological and field conditions allow for operational adjustments in the Digital Twin of the field.

- MLE improves back allocation accuracy at well and depth level.
- Integration into Digital Twin enables material balance consistency and operational corrections.
- EOS + CG + PLT + PVT together provide sufficient input for real-time allocation.
- Approach yields better reserve attribution, reduced risk, improved field development strategy.

Way Forward

- Extend MLE-based back allocation to **full-field deployment** in large-scale Digital Twin projects.
- Incorporate **AI/ML models** for predictive calibration of EOS/CG in real time.
- Test applicability in **enhanced recovery schemes** with gas injection and compositional changes.
- Benchmark performance across multiple basins to standardize adoption.

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Nomenclature

CCE	Constant Composition Expansion
CVD	Constant Volume Depletion
CG	Compositional Grading

CPF Central Processing Facilities
EOS Equation of State
MLE Maximum Likelihood Estimation
PVT Pressure-Volume-Temperature
PLT Production Logging Tool

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